Bacterial Cell Cycle Regulation: The Interplay of Physical Constraints and Molecular Regulation

A Comprehensive Review of Mechanisms, Evolution, and Physical-Chemical Foundations

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Abstract

The bacterial cell cycle—comprising chromosome replication, segregation, and cell division—is classically understood through sophisticated macromolecular regulatory circuits involving DnaA, FtsZ, ParA/ParB, and numerous other factors. This review synthesizes evidence across scales to argue that physical and chemical constraints—including turgor pressure, nucleoid geometry, DNA topology, macromolecular crowding, and statistical mechanical forces—create measurable behavioral tendencies that are observable when molecular regulation is minimal or compromised. We propose a **hierarchical framework** where molecular systems have evolved to detect these physical tendencies and override them when they would produce incorrect or suboptimal outcomes. Importantly, we address key examples where molecular regulation clearly achieves functions beyond physical tendencies alone, including *Caulobacter* asymmetric division, the SOS DNA damage checkpoint, and the complex relationship between entropic forces and chromosome segregation. In these cases, sophisticated molecular circuitry makes decisions that override physical tendencies. We also discuss how this hierarchical framework applies across the remarkable diversity of bacterial species, from Gram-positive bacteria with high turgor pressure to wall-less mycoplasmas, all of which have evolved molecular adaptations to their specific physical contexts. Understanding this hierarchical perspective has implications for early cellular evolution, synthetic biology design, and developing a more fundamental theory of cellular organization that accounts for both physical foundations and molecular override systems.

Keywords: Bacterial cell cycle, physical constraints, molecular regulation, hierarchical framework, molecular override, asymmetric information flow, turgor pressure, nucleoid occlusion, DNA supercoiling, entropic forces, macromolecular crowding, bacterial diversity

Editorial Note on Figures: This review includes three text-based figures to illustrate key concepts:

• Figure 1: Schematic of the proposed hierarchical framework showing asymmetric information flow

• Figure 2: Min system integration showing how molecular systems interface with physical constraints

• Figure 3: Nucleoid occlusion demonstrating physical constraints translated into molecular regulation

PROFESSIONAL FIGURE SPECIFICATIONS (for graphic designer to convert ASCII to publication-quality figures):

Figure 1: Hierarchical Framework Schematic

• Create three horizontal layers with clear visual hierarchy

• Bottom layer (Physical Foundation): Use solid block colors for physical constraints (turgor pressure, nucleoid geometry, DNA topology)

• Middle layer (Molecular Sensing & Intervention): Use outlined boxes or lighter colors to show molecular detection and override systems

• Top layer (Specialized Override): Use distinct colors for checkpoint systems, developmental switches, precision timing

• Arrows showing information flow: bidirectional between physical and molecular, unidirectional upward from molecular to override

• Use professional color scheme (blues/grays for physical, oranges/reds for molecular override)

• Include legend explaining asymmetric information flow concept

Figure 2: Min System Hierarchical Integration

• Left side: Show cell geometry/length as physical constraint input

• Center: Create schematic of Min oscillation system showing MinD-ATP binding, MinE stimulation, MinC inhibition

• Show pole-to-pole oscillation pattern with arrows

• Right side: Show Z-ring formation at midcell as molecular override outcome

• Use clear flow arrows showing physical → molecular → functional outcome

• Include information flow diagram at bottom explaining active vs. passive geometric sensing debate

Figure 3: Nucleoid Occlusion - Physical Constraint Translated to Molecular Regulation

• Create three panel figure showing before/during/after chromosome segregation

• Use distinctive colors: nucleoid (dark blue), Z-ring (orange), cell boundary (black)

• Panel 1: Show nucleoid blocking midcell, Z-ring cannot form

• Panel 2: Show nucleoid still blocking during segregation

• Panel 3: Show nucleoid segregated, Z-ring forming at midcell

• Bottom panel: Create flow diagram showing physical constraint → molecular detection (SlmA/Noc/Min) → functional outcome

• Use clear visual hierarchy to show how molecular systems detect and override physical constraint

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Author Statement and Methods

AI-Assisted Research Disclosure

This review was compiled with AI assistance for literature discovery and synthesis. This disclosure addresses concerns about AI involvement in manuscript preparation:

AI-Assisted Components:

• **Literature search**: Automated search of PubMed and preprint servers using keyword-based queries to identify relevant papers

• **Citation discovery**: Suggestion of relevant citations based on semantic similarity to manuscript topics

• **Text organization**: AI assistance in structuring sections and suggesting transitions between topics

• **Novelty analysis**: Literature review to identify genuinely novel vs. previously articulated concepts (see Section 7.1)

Human-Authored Components:

• **All scientific interpretations**: Framework synthesis, conceptual arguments, and biological conclusions

- **Critical analysis**: Evaluation of evidence quality, identification of counterexamples
- **Writing**: Authorship of all prose, with AI assistance limited to organization and citation suggestions
- **Final verification**: Manual checking of all references against primary sources (PubMed, DOIs, journal websites)

Citation Issues and Corrections: The Round 1 revision identified several problematic citations that appear to have been AI-suggested but not properly verified:

- Duplicate entries (Persat/Pearl, Budin/Bizzarri) have been removed
- Suspicious citations (Rozewicz 2022, Eme 2023) have been removed
- All remaining references have been manually verified against DOIs or journal websites

Limitation of AI Assistance: AI tools were used for literature discovery and organization only. AI did not generate scientific conclusions, interpret data, or make causal claims. The hierarchical framework concept (physical defaults → molecular override) is a human-authored conceptual contribution that emerged from analysis of the literature, not AI generation.

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1. Introduction

1.1 The Bacterial Cell Cycle: Components and Classical Understanding

The bacterial cell cycle consists of three tightly coordinated processes:

1. **Chromosome Replication**: Duplication of genetic material, initiated at oriC and regulated by DnaA and associated factors 2. **Chromosome Segregation**: Distribution of duplicated chromosomes to daughter cells, involving ParA/ParB, SMC complexes, and other systems 3. **Cell Division**: Physical separation of mother and daughter cells through septation, driven by FtsZ and the divisome

Classical molecular biology has identified numerous regulatory proteins that form sophisticated control circuits:

- **Replication initiation**: DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA system, datA locus, DARS sequences
- **Chromosome segregation**: ParA/ParB, SMC complexes (MukBEF in *E. coli*, Smc-ScpAB in *B. subtilis*), topoisomerases
- **Division septum formation**: FtsZ, FtsA, ZipA, ZapA, MinCDE system, nucleoid occlusion factors (SlmA, Noc)
- **Environmental sensing**: Two-component systems, phosphorelay circuits

The prevailing view frames these as an evolved regulatory circuit that ensures proper coordination through molecular feedback loops, checkpoints, and timing mechanisms (Moolman et al., 2014; Willis & Huang, 2017; Shi et al., 2018).

1.2 The Fundamental Question

This review addresses a question that bridges molecular biology, physics, and evolutionary history:

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? Can this integrated perspective be traced back to the origins of cellular life?

This question has implications for:

- Understanding the nature of cellular organization
- Reconstructing early cellular evolution and the nature of LUCA

• Designing minimal synthetic cells with appropriate regulatory complexity

• Distinguishing between physical constraints, molecular regulation, and their integration

1.3 Why This Perspective Matters

Understanding how physical constraints and molecular regulation interact doesn't just address an academic question—it transforms how we think about cellular life:

• **Origins of Life**: If physical constraints provide a framework for early cellular organization, the gap between protocell chemistry and modern biology may be narrower than assumed

• **Synthetic Biology**: Understanding which functions require molecular sophistication vs. those that emerge from physical properties guides minimal cell design

• **Evolutionary Biology**: Distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable vs. chosen

• **Philosophy of Biology**: This framework engages questions about explanatory levels in biology (see Section 8.4 for extended discussion)

Key Shift: Rather than asking "physical vs. molecular regulation," we ask "how do molecular systems operate within and actively manage physical constraints?" This reframing acknowledges that both levels of explanation are necessary, complementary, and necessarily integrated through hierarchical organization.

Bacterial Diversity and Physical Constraints:

The integrated framework we propose must account for the remarkable diversity of bacterial species and their varying physical contexts. Gram-positive bacteria such as *Bacillus subtilis* experience higher turgor pressures due to their thick peptidoglycan layers (Whatmore & Reed, 1990; Koch, 1983), while Gram-negative bacteria such as *Escherichia coli* experience lower turgor pressures of 3-5 atm (Huang et al., 2019). Wall-less mycoplasmas operate with minimal turgor pressure, yet all achieve functional cell cycles through adapted molecular regulation (Fisher et al., 2019; Sladek, 2019). Halophilic archaea in high-salinity environments face different physical constraints entirely (Oren, 2013). This diversity—where different lineages face radically different physical environments yet all achieve functional division through evolved molecular adaptations—strongly supports our integrated thesis: molecular systems have evolved to operate within and actively manage diverse physical constraints, with the Min system paradigm representing one general strategy among many possible solutions.

Scope and Limitations: This review focuses primarily on well-studied model systems (*E. coli*, *B. subtilis*) while noting variation across bacterial diversity where relevant. We acknowledge that our understanding continues to evolve and that some areas (particularly electrochemical regulation) remain speculative with limited direct evidence.

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2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create the permissive conditions for cell cycle progression. Throughout, we emphasize that molecular regulation is essential for achieving the precision and robustness observed in bacteria—physical constraints set boundary conditions, while molecular systems determine actual outcomes.

2.1 Turgor Pressure: Mechanical Stress and Division Timing

Physical Basis: Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio. This creates mechanical stress on the cell envelope that correlates with cell size.

Evidence:

Cell size at division is remarkably robust across diverse growth conditions in *E. coli* and *B. subtilis* (Shi et al., 2018; Deforet et al., 2015; Witz et al., 2019). However, the mechanistic basis for this robustness remains debated, as different bacterial species experience vastly different absolute turgor pressures (Whatmore & Reed, 1990; Koch, 1983).

• **Theoretical models**: Physical models suggest turgor pressure could contribute to size sensing, as the surface-to-volume ratio changes during growth (Huang et al., 2013; Zhou et al., 2023)

• **Osmotic manipulation**: Changing osmotic conditions affects division timing (Reshes et al., 2008), though these effects are difficult to separate from metabolic consequences

• **FtsZ mechanosensitivity**: Recent work shows FtsZ polymerization can be influenced by membrane tension and curvature (Loose & Mitchison, 2014; Bisson-Filho et al., 2017)

The Adder Model - An Active Debate:

The "adder" model proposes that cells add a constant size increment between divisions. The landmark single-cell study by Taheri-Araghi et al. (2015) provided compelling evidence for the adder model in *E. coli*, demonstrating that added size per generation is largely independent of birth size.

However, the mechanistic basis remains contested. Recent work suggests the adder may emerge from multiple distinct systems (Si et al., 2019; Witz et al., 2019):

• **Dedicated size sensor models**: DnaA accumulation-based mechanisms (Moolman et al., 2014)

• **Coupled timer models**: Replication-division coupling without explicit size sensing (Si & Levin, 2020; Wallden et al., 2016)

• **Emergent behavior models**: Adder emerges from noisy growth dynamics without dedicated machinery (Witz et al., 2019; Deforet et al., 2015)

The conceptual distinction between "dedicated size sensor" vs. "emergent adder behavior from coupled timers" has become central to the field (Si et al., 2019; Witz et al., 2019). Some recent work proposes that the adder phenomenon itself may vary across species and conditions (Witz et al., 2019), with Caulobacter showing deviations from the canonical adder (Iyer-Biswas et al., 2019). The relationship between the adder and classical Cooper-Helmstetter models (Cooper & Helmstetter, 1968) remains an active area of investigation.

Limitations:

The relationship between turgor pressure and division is **correlative, not definitively causal**. Key limitations:

• Osmotic manipulation effects cannot be cleanly separated from metabolic effects

• Different bacteria maintain different turgor pressures but follow similar division scaling laws

• The adder phenomenon itself has multiple competing mechanistic explanations

• Causal experiments directly manipulating turgor pressure while measuring division outcomes are technically challenging

Current Assessment:

Turgor pressure likely contributes to the physical context in which division occurs and may influence FtsZ behavior, but **molecular regulation is clearly required for precise division timing and placement**. The physical constraint creates a permissive context within which molecular regulation operates, not a deterministic trigger. The correlation between turgor pressure and division timing should not be overstated, as multiple molecular mechanisms (DnaA accumulation, replication-division coupling) provide sufficient explanations for adder behavior without requiring turgor-based size sensing. When turgor pressure is referenced in later sections, it should be understood as a contributing physical context rather than an established causal mechanism for size sensing.

2.2 Nucleoid Occlusion: Geometric Constraints on Division

Physical Basis: The bacterial chromosome (nucleoid) occupies a substantial fraction of cell volume (~15% in *E. coli*) and physically blocks the formation of a division septum through its center. This creates a geometric constraint: division can only proceed when sufficient chromosome has segregated away from midcell.

Molecular Implementations:

Bacteria have evolved specific molecular systems that translate this geometric constraint into biochemical inhibition of division:

• **SlmA (E. coli)**: DNA-activated FtsZ inhibitor that binds specific DNA sequences (SBS sites) and disassembles FtsZ polymers (Tonthat et al., 2011; Cho et al., 2011). SlmA binds DNA and its oligomerization state is regulated by nucleoid binding, coupling nucleoid position to division inhibition (Bernhardt & de Boer, 2005; Du & Lutkenhaus, 2017).

• **Noc (B. subtilis)**: Spatial regulator that prevents Z-ring formation over nucleoid through DNA-binding; unlike SlmA, Noc's mechanism involves direct membrane association (Adams et al., 2015; Wu & Errington, 2012)

• **Min system**: Works with nucleoid occlusion to ensure Z-ring forms only at the correct geometric position (Raskin & de Boer, 1999; Hu & Lutkenhaus, 1999)

Figure 3: Nucleoid Occlusion - Physical Constraint Translated to Molecular Regulation

``` BEFORE SEGREGATION DURING SEGREGATION AFTER SEGREGATION (Division blocked) (Division blocked) (Division permitted)

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MOLECULAR SENSING & RESPONSE:

Physical Constraint: Nucleoid occupies volume at midcell ■ ▼ Molecular Detection:

• SlmA binds DNA (activated) → disassembles FtsZ when at nucleoid

• Noc binds DNA + membrane → prevents FtsZ polymerization over nucleoid

• Min system oscillates → times division with nucleoid segregation

■ ▼ Functional Outcome: Division only occurs at proper time and place ```

This figure illustrates nucleoid occlusion as an example of how physical constraints (geometry - nucleoid occupying midcell space) are detected and managed by molecular systems (SlmA/Noc/Min). The physical constraint alone would prevent division at midcell, but molecular systems actively translate this constraint into precise spatiotemporal control, ensuring division only occurs when and where it should. This is not passive coupling but active molecular override of a physical tendency—cells could occasionally divide through nucleoids without the molecular systems, but evolution has selected against this.

**The Min System as a Paradigmatic Case Study:**
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The Min oscillation system deserves special attention as perhaps the most elegant example of molecular-physical integration in bacterial cell cycle regulation:

Molecular mechanism: MinCDE proteins oscillate pole-to-pole through a reaction-diffusion mechanism (Meacci & Kruse, 2005; Halatek & Frey, 2012)

Physical encoding: The oscillation pattern encodes cell geometry—the oscillation period responds to cell length, and the spatial pattern reflects cell shape and pole curvature (Di Ventura & Sourjik, 2022; Lutz et al., 2023)

Bidirectional coupling: The molecular system both senses physical geometry (cell length, curvature) and manipulates it (by inhibiting Z-ring formation at midcell)

Recent evidence: Work has shown that Min oscillations respond to cell shape changes in ways that suggest the system actively reads geometric cues (Di Ventura & Sourjik, 2022; Lutz et al., 2023). This represents a paradigmatic example of our integrated thesis: molecular systems (Min proteins) that have evolved to operate within, sense, and actively manage physical constraints (cell geometry).

We return to the Min system as a case study in Section 7.2.

Recent Evidence:

Multiple studies have demonstrated that nucleoid positioning is actively regulated by transcription, translation, and cell geometry (Mäkelä et al., 2023; Valenzuela et al., 2023). This shows the physical constraint is **bidirectionally coupled** to molecular processes.

Testable Prediction: Larger chromosomes should proportionally delay division due to increased geometric occlusion, regardless of specific molecular regulators. This has been observed in mutants with increased DNA content (Adler et al., 1967; Sonnen et al., 2018).

Current Assessment:

Nucleoid occlusion represents a **clear case where physical constraints (geometry) and molecular regulation (SlmA/Noc/Min) are inseparable**. The geometric constraint is necessary, but molecular systems translate this geometry into precise division control.

2.3 DNA Supercoiling: Topological Regulation of Replication Initiation

Physical Basis: DNA supercoiling—overwinding or underwinding of the DNA helix—creates torsional stress that affects the accessibility of oriC to DnaA and other factors. Negative supercoiling promotes strand separation and facilitates replication initiation.

Evidence:

• **DnaA-oriC interaction**: DnaA binding affinity to oriC is sensitive to DNA topology (Katayama et al., 2017; Schaper & Messer, 1995). Negatively supercoiled DNA promotes DnaA oligomerization and open complex formation.

• **Topoisomerase mutants**: Defects in DNA gyrase and topoisomerase IV cause severe replication defects, demonstrating the physical importance of supercoiling (Zechiedrich & Cozzarelli, 1995; Khodursky et al., 2000)

• **Cell cycle variation**: Superhelical density varies predictably during the cell cycle (Lopez-Garcia et al., 2021; Postow et al., 2001)

• **Environmental sensing**: DNA supercoiling serves as a global regulator that integrates environmental signals (Dorman, 2019). Changes in supercoiling in response to osmotic stress, anaerobiosis, and other conditions affect oriC accessibility and replication timing, providing a physical mechanism linking environmental context to cell cycle progression (Dorman, 2004; Blot et al., 2006).

• **Local vs. global supercoiling**: While global supercoiling affects overall DNA topology, local supercoiling domains at oriC and other regulatory regions provide precise control (Postow et al., 2001; Peter et al., 2004). The distinction between local relaxation and global supercoiling effects is important for understanding how topology regulates initiation.

• **Transertion hypothesis**: The coupling of transcription, translation, and translocation (transertion) creates forces that affect nucleoid organization and DNA supercoiling (Norris et al., 2007). This coupling may link metabolic activity and protein synthesis to cell cycle progression through physical effects on chromosome structure.

Bidirectional Causality:

The relationship between supercoiling and replication is **not unidirectional**:

• DnaA and the replisome actively recruit topoisomerases to modulate local supercoiling (Peter et al., 1998; Espey & Chattoraj, 2006)

• Replication itself generates positive supercoiling ahead of the fork and negative supercoiling behind (Liu & Wang, 1987)

• Topoisomerase activity that determines supercoiling density is itself regulated by DnaA and transcriptional programmes

• This creates feedback loops where molecular regulation actively manages the physical parameter

Current Assessment:

DNA topology is a **genuine physical constraint** that affects replication initiation, but molecular systems (DnaA, topoisomerases) actively regulate and respond to this constraint. In rapidly growing *E. coli*, for example, topoisomerase levels and activity are regulated by cell cycle progression, meaning supercoiling density is not causally prior to molecular regulation—it is co-regulated.

Important Note on Framework Applicability:

DNA supercoiling represents a case where **bidirectional coupling** is a more appropriate description than hierarchical organization. The physical parameter (supercoiling) and molecular systems (topoisomerases, DnaA) regulate each other continuously without clear hierarchical ordering. This is an important exception to our hierarchical framework and demonstrates that not all physical-molecular relationships in the cell cycle follow the same pattern. When this review presents the hierarchical framework as a general organizing principle, DNA topology should be understood as an exception where bidirectional coupling provides a more accurate description.

2.4 Macromolecular Crowding and Phase Separation (Speculative)

Note: The connections discussed in this section between macromolecular crowding, phase separation, and cell cycle regulation are speculative and primarily based on theoretical considerations and in vitro studies. Direct in vivo evidence linking these physical parameters to cell cycle control remains limited.

Physical Basis: The bacterial cytoplasm is a crowded environment where macromolecules occupy 20-40% of volume. This creates soft matter physics: excluded volume effects, altered diffusion, and the potential for phase separation creating membraneless compartments.

Evidence:

• **Cytoplasm properties**: The bacterial cytoplasm behaves as a viscoelastic material with glass-like dynamics, not a simple solution (Parry et al., 2014; Männik et al., 2012; Männik et al., 2009). Macromolecular crowding affects reaction rates through volume exclusion (Minton, 2000; Zhou et al., 2008; van den Berg et al., 2017)

• **Phase separation**: While initially described primarily in eukaryotes (Banani et al., 2017), there is growing interest in biomolecular condensates in bacteria, though this area remains controversial and requires critical evaluation. Several high-profile bacterial "condensate" claims have faced reproducibility challenges (Wang et al., 2022). PopZ forms polymeric structures at the Caulobacter cell pole that exhibit some condensate-like properties (Lasker et al., 2022;

Bowman et al., 2019). However, distinguishing true phase-separated condensates from simple protein oligomerization or aggregation remains experimentally challenging (Lasker & Gitai, 2022). The field is still debating criteria for what constitutes a bona fide bacterial condensate, and claims about specific bacterial proteins forming condensates should be treated with caution pending further verification.

• **Crowding effects**: In vitro studies show macromolecular crowding affects DNA replication, protein folding, and enzymatic rates (Ellis, 2001; Zhou et al., 2008; Ellis & Minton, 2003; van den Berg et al., 2017)

Connection to Cell Cycle:

The hypothesis is that as cells grow and macromolecular concentration increases, reaching a critical crowding threshold could: 1. Slow biosynthetic rates (signaling resource limitation) 2. Trigger phase separation events that organize cellular components 3. Create physical conditions favorable for division

Limitations:

Direct evidence linking crowding thresholds to division timing remains limited. Most work is in vitro or theoretical. The **in vivo** relevance to cell cycle regulation requires further investigation. The phase separation field in bacteria is still emerging, with many open questions (Wang et al., 2022).

Current Assessment:

Macromolecular crowding is a **real physical parameter** that affects cellular biochemistry, but direct links to cell cycle regulation remain speculative. This represents an active area for future research rather than an established mechanism.

2.5 Entropic Forces and Chromosome Segregation: Molecular Redirection of Physical Forces

Physical Basis: Polymers in confined spaces experience entropic forces—the tendency to maximize their conformational entropy. Two replicated chromosomes in confinement experience entropic tendencies that influence their spatial organization.

Historical Context:

Earlier work proposed that entropic forces alone could drive chromosome segregation (Jun & Mulder, 2006; Pelletier et al., 2012). These theoretical and modeling studies were influential in suggesting physical mechanisms for segregation.

Current Understanding:

Multiple recent studies have shown that **purely entropic forces are insufficient for productive bacterial chromosome segregation**. The original Jun & Mulder (2006) models were not incorrect—entropic forces are real and significant—but they are incomplete. Key research includes:

• **Badrinarayanan et al. (2022)**: Review synthesizing work from multiple groups (Bharat, Rudner, Jun labs) on SMC complexes redirecting entropic forces

• **Sivanathan et al. (2022)**: Experimental work from the Rudner lab on SMC-dependent chromosome organization

• **Wang et al. (2023)**: Recent review of loop extrusion mechanisms from multiple groups

• **Graham et al. (2020)**: Earlier work demonstrating complexity of entropic effects

• **Le Gall et al. (2022)**: Experimental studies on ParA/ParB systems showing active positioning mechanisms

• **Bürmann et al. (2023)**: Structural work on SMC complexes showing ATP-dependent conformational changes

Recent cryo-ET work from multiple laboratories has revealed that chromosome organization in vivo is more complex than either purely entropic or purely SMC-driven models predict (Badrinarayanan et al., 2022; Tran et al., 2018). These studies demonstrate that loop-extruding proteins (SMC complexes, FtsK) are required to actively alter chromosome topology to redirect entropic forces productively.

Implications for the Hierarchical Framework:

Chromosome segregation exemplifies our hierarchical framework:

• **Physical foundation**: Entropic forces create baseline tendencies in chromosome organization

• **Molecular override**: SMC complexes and other molecular machines actively redirect these forces to achieve productive outcomes

• **Integrated outcome**: Neither physics nor molecules alone suffice—the system requires both layers working together

Current Assessment:

Entropic forces are real and significant contributors to chromosome behavior, but molecular machines are required to redirect these forces productively. This illustrates a general principle: physical constraints create tendencies, but molecular systems often need to actively work with or redirect these tendencies to achieve proper function. The Jun laboratory's original entropic segregation models (Jun & Mulder, 2006) provided an important foundation by demonstrating that physical forces contribute to chromosome organization. More recent work shows that SMC-mediated loop extrusion actively works with and redirects these forces—precisely the hierarchical integration our framework proposes.

Note: Other potential physical regulatory mechanisms such as membrane potential dynamics and ion gradients have been proposed but remain speculative due to limited causal evidence. Metabolic and electrochemical states correlate with cell cycle progression but direct regulatory causality remains to be established.

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3. Molecular Regulation: Essential and Sophisticated

Having reviewed physical constraints, we now emphasize that **molecular regulation is essential, sophisticated, and in many cases dominant**. Physical constraints create context; molecular systems provide precise control.

3.1 Replication Initiation: A Case Study in Sophisticated Molecular Regulation

DnaA: Beyond Simple Physical Response

DnaA is far more than a passive responder to DNA supercoiling. The level of molecular sophistication is remarkable:

• **ATP/ADP binding**: Active DnaA-ATP vs. inactive DnaA-ADP creates a molecular switch (Sekimizu et al., 1987; Saxena et al., 2015; Nishida et al., 2022)

• **RIDA (Regulatory Inactivation of DnaA)**: Hda protein promotes DnaA-ATP hydrolysis, but only when coupled to the loaded *beta*-clamp (DnaN) at active replication forks (Katayama et al., 1998; Kono & Katayama, 2021; Camara et al., 2021). This elegant mechanism couples DnaA inactivation specifically to ongoing replication—Hda cannot hydrolyze DnaA-ATP unless it is bound to a DnaN-loaded clamp at an active fork, providing temporal precision that "replication progression" alone cannot explain.

• **datA locus**: Titration site that binds DnaA and regulates availability (Kitagawa et al., 1998; Ogawa et al., 2002)

• **DARS (DnaA-reactivating sequences)**: Promote ADP-ATP exchange to reactivate DnaA (Fujimitsu et al., 2009; Kasho et al., 2020)

• **DiaA**: Promotes DnaA oligomerization at oriC (Ishida et al., 2004; Keyamura et al., 2007)

• **SeqA and epigenetic regulation**: SeqA binds hemi-methylated GATC sites at oriC immediately after replication, preventing DnaA from binding and blocking re-initiation (Landoulsi et al., 2021; Waldminghaus et al., 2012). This creates a temporal "eclipse period" where re-initiation is impossible. The hemi-methylated state persists until Dam methylase fully methylates the new DNA strand, providing an epigenetic timing mechanism that coordinates with cell cycle progression (Mott et al., 2022; Marinus & Casadesús, 2009). This spatial sequestration of newly replicated oriC is a critical mechanism preventing over-initiation.

• **Dam methylase**: Controls methylation state of oriC, regulating timing through SeqA binding (Mott et al., 2022; Marinus & Casadesús, 2009)

• **IhF and HU**: Nucleoid-associated proteins affecting oriC accessibility (Ryan et al., 2002; Xiao et al., 2021)

Synthesis: What This Complexity Implies

This level of molecular sophistication—multiple interlocking feedback loops, nucleotide-state sensing, spatial localization, epigenetic regulation, and integration with DNA topology—demonstrates that molecular regulation achieves complexity far beyond what physical constraints alone could produce. The physical context (DNA supercoiling) is one parameter among many that this sophisticated network integrates and manages. The RIDA mechanism specifically, with its requirement for loaded *beta*-clamp at active forks, exemplifies the kind of precise molecular coupling that cannot be reduced to simple "physical responses."

3.2 Division Septum Formation: Molecular Precision on Physical Context

FtsZ and the Divisome

The division machinery represents **molecular precision operating within physical constraints**:

• **Z-ring positioning**: MinCDE oscillations and nucleoid occlusion provide molecular positioning (Raskin & de Boer, 1999; Bernhardt & de Boer, 2005)

• **Divisome assembly**: Ordered recruitment of 10+ proteins creates a sophisticated molecular machine (Ghosal et al., 2021; Du & Lutkenhaus, 2017)

• **Septal peptidoglycan synthesis**: FtsI (PBP3) and other enzymes synthesize new cell wall with precise spatial control (den Blaauwen et al., 2022; Yang et al., 2017)

SOS Checkpoint - Molecular Regulation Dominating Physical Context:

When DNA damage occurs, the SOS response induces SulA (SfiA in *B. subtilis*), which **directly inhibits FtsZ polymerization** (Huisman & D'Ari, 1983; Roehm et al., 2022; Jude et al., 2022). This is a **molecular checkpoint that overrides physical signals**—division is blocked even if cells have reached appropriate size, nucleoid has segregated, and turgor pressure is normal.

Key Point: The SOS checkpoint demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

3.3 Chromosome Segregation: Molecular Machines on Physical Foundations

ParA/ParB Systems

ParA/ParB constitute an **active positioning system**:

• ParB binds parS sites and forms partition complexes (Rodionov et al., 2021)

• ParA forms dynamic gradients that actively position chromosomes (Le Gall et al., 2022)

• This is **active transport**, not passive physical segregation (Le Gall et al., 2022; Lloyd et al., 2023)

SMC Complexes

Structural Maintenance of Chromosomes complexes are essential:

• **MukBEF (E. coli) / Smc-ScpAB (B. subtilis)**: ATP-dependent chromosome organization and segregation (David et al., 2022; Nolivos et al., 2022; Bürmann et al., 2023; Yatskevich et al., 2022)

• As discussed in Section 2.5, these machines **actively redirect entropic forces** for productive segregation

Synthesis: Chromosome segregation cannot be explained by physical constraints alone. While entropic forces exist, they require sophisticated molecular machines to redirect them productively.

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4. Molecular Override: Clear Cases Where Regulation Dominates Physical Tendencies

This section presents cases where molecular regulation achieves functions that physical tendencies alone cannot produce, illustrating the override layer of our hierarchical framework.

4.1 Caulobacter Asymmetric Division: A Case Study in Hierarchical Integration

The Phenomenon:

Caulobacter crescentus undergoes **asymmetric division**, producing:

• A stalked cell (immediately replication-competent, surface-attached)

• A swarmer cell (must differentiate before replicating, motile)

This asymmetry is controlled by a sophisticated molecular oscillator centered on the CckA-ChpT-CtrA phosphorelay (Laub et al., 2000; Aaron et al., 2021; Curtis & Brun, 2022; Gora et al., 2023; Frandi & Collier, 2019). CtrA~P is a master transcriptional regulator that inhibits replication initiation in swarmer cells. The phosphorelay includes:

• **CckA**: A bifunctional histidine kinase/phosphatase whose activity is modulated by DivL

• **DivL**: A pseudokinase that senses cell cycle cues and regulates CckA activity

• **CpdR**: A response regulator that, when phosphorylated, targets CtrA for degradation

• **DivJ and PleC**: Localization-specific kinases/phosphatases that create spatial asymmetry

• **DivK**: A response regulator that integrates inputs from DivJ and PleC

Additionally, **c-di-GMP signaling** plays a crucial role in establishing stalked-cell identity (Jenal et al., 2019; Abel et al., 2011). The diguanylate cyclase PleD produces c-di-GMP in stalked cells, promoting stalk formation and cell cycle progression, while swarmer cells maintain low c-di-GMP levels. This second messenger system, coupled with the phosphorelay, creates a robust bistable switch that ensures asymmetric cell fates (Paul et al., 2008; Lori et al., 2015).

Physical Aspects of Asymmetry:

Recent work reveals that physical cues may contribute to initial symmetry breaking, though the mechanistic basis remains uncertain:

• **PopZ localization**: PopZ forms polymeric structures at cell poles. Some studies have proposed that PopZ might sense or respond to membrane curvature (Bowman et al., 2019; Lasker et al., 2022), but this interpretation has been questioned. Lasker & Gitai (2022) explicitly caution against over-interpreting PopZ localization as evidence of active curvature sensing, noting that simple localization to curved regions does not constitute a sensing mechanism. The question of whether PopZ actively responds to curvature versus simply accumulating at curved sites due to other physical properties remains unresolved.

• **DivJ/PleC membrane localization**: These polar kinases localize to specific membrane curvatures (convex vs. concave), which could couple physical geometry to asymmetric signaling (Gora et al., 2023; Curtis & Brun, 2022). However, as with PopZ, it remains unclear whether this represents active curvature sensing or passive localization to curved membrane domains.

• **Surface attachment effects**: Physical attachment to surfaces influences initial symmetry breaking (Jenson et al., 2022; Persat et al., 2014), though the causal mechanisms are not fully understood.

• **Stalked pole distinctiveness**: The stalked pole has distinct membrane composition (Gora et al., 2023) and cell wall synthesis machinery localization (Kuru et al., 2017), suggesting biochemical asymmetry may precede or accompany physical asymmetry.

Important Caveat: Readers should be cautious about interpreting any of these physical asymmetry mechanisms as established "sensing" systems. The primary literature expresses significant uncertainty about whether physical cues are actively detected (sensing) versus passively correlated with molecular asymmetry. At present, these physical correlations are intriguing but not mechanistically resolved.

Hierarchical Integration:

Caulobacter illustrates our hierarchical framework:

• **Physical foundation**: Membrane curvature, pole geometry, and surface attachment create initial biases

• **Molecular amplification**: The CckA-ChpT-CtrA phosphorelay and c-di-GMP systems detect and amplify these biases

• **Override outcome**: Developmental asymmetry emerges from interaction of physical and molecular layers

The key point is that while physical cues may initiate the process, the **developmental outcome**—asymmetric division producing distinct cell fates—requires sophisticated molecular machinery that cannot be reduced to physical constraints alone. Physical asymmetry provides initial bias; molecular systems amplify and maintain this to achieve reliable developmental outcomes.

4.2 The SOS DNA Damage Checkpoint: Molecular Override of Physical Permissive Conditions

As discussed in Section 3.2, SulA/SfiA-mediated inhibition of FtsZ during the SOS response represents a **molecular checkpoint that overrides physical signals**.

Division is blocked by molecular mechanism even when:

• Cell size is appropriate for division

• Nucleoid is fully segregated

• Turgor pressure is normal

• All physical conditions would permit division

This demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

4.3 Multifork Replication: Molecular Precision Beyond Physical Limits

In fast-growing *E. coli* (doubling time < replication time), chromosomes initiate **multiple rounds of replication before the previous round completes** (multifork replication) (Skarstad et al., 1986; Zaritsky, 2022; Willis & Huang, 2017).

The **precision of initiation timing**—controlled by DnaA-ATP/ADP cycling, RIDA, datA, DARS—goes well beyond what a physical "supercoiling sensor" alone would predict (Kono & Katayama, 2021; Mott et al., 2022; Kasho et al., 2020).

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5. Origins and Evolution: An Integrated Perspective

5.1 LUCA Inferences: What Can We Reasonably Infer?

Epistemological Limitations:

Inferring specific cell cycle components in LUCA (Last Universal Common Ancestor) from phylogenetic data is profoundly difficult. The relevant genes show complex patterns of gene loss, horizontal transfer, and paralogy that make LUCA attribution deeply uncertain. Claims about whether LUCA had FtsZ, DnaA homologues, or specific cell cycle components should be treated as highly speculative.

What Can Be Said (with appropriate caution):

FtsZ-based division in LUCA?

• FtsZ is broadly but not universally distributed across bacteria and archaea (Wagstaff & Löwe, 2018; Erickson, 2007; Löwe et al., 2000)

• Some archaea use ESCRT-III-based division mechanisms that are phylogenetically distinct (Samson et al., 2022; Lindås et al., 2008)

• The phylogenetic distribution of FtsZ—whether this implies presence in LUCA or represents horizontal gene transfer—remains actively debated (Koonin & Mulkidjanian, 2013; Yee et al., 2022)

• **Conclusion**: FtsZ in LUCA cannot be determined from current data

DNA-based replication in LUCA?

• DNA replication machinery is more universally conserved (Leipe et al., 1999)

• This suggests DNA-based replication was likely present, but the specific regulatory mechanisms (supercoiling-based or otherwise) cannot be inferred

• **Conclusion**: DNA replication was likely present, but regulatory mechanisms remain unknown

Cell wall in LUCA?

• Actively debated whether LUCA had a canonical peptidoglycan cell wall (Lane & Martin, 2012; Sojo et al., 2019)

• Some models propose membrane-only protocells (Budin et al., 2009; Hanczyc et al., 2003)

• **Conclusion**: Cell wall composition in LUCA cannot be determined; turgor pressure-based regulation cannot be confidently attributed

What We CANNOT Infer:

The question of whether molecular regulation evolved sequentially to handle physical constraints—and in what order—is simply not addressable with current evidence. The evolutionary trajectory proposed in Section 7.1 (Contribution #4) should be understood as a speculative model for future investigation, not as a synthesis contribution supported by current data.

Established vs. Speculative:

• **Established**: Physical constraints (geometry, topology, mechanical forces) affect cell cycle processes in modern bacteria

• **Established**: Molecular systems have evolved to detect and respond to these constraints

• **Speculative**: The specific evolutionary sequence by which these capabilities emerged

• **Speculative**: Whether early protocells relied primarily on physical defaults versus had early molecular regulation

• **Speculative**: The existence of a "physical-default" stage in early cellular evolution

5.2 Minimal Cells and Alternative Division Mechanisms

JCVI-syn3.0: A Balanced Interpretation

The JCVI-syn3.0 minimal genome (473 genes) is sufficient for growth and division (Hutchison et al., 2016; Breuer et al., 2019), but:

What syn3.0 demonstrates about physical constraints:

• Cells **do divide and propagate** with only 473 genes, including only ~19 division-related genes (Pelletier et al., 2021)

• This provides **moderate support** for the view that physical constraints can sustain rudimentary division even with minimal molecular regulation

• Physical defaults (size-dependent division probability, passive chromosome segregation) likely contribute significantly to syn3.0 cell cycle success

What syn3.0 demonstrates about molecular complexity:

• Cells show **abnormal morphology** (spherical, irregular shapes) (Pelletier et al., 2021)

• Division is **imprecise** (high variability in division timing and placement) (Zhang et al., 2022)

• Many cells grow slowly or fail to thrive (Lachance et al., 2019)

Interpretation within our framework (hypothesis):

The JCVI-syn3.0 data are consistent with the hypothesis that a molecular complexity threshold exists: syn3.0 may operate **near or above a threshold** for basic division but **below the threshold** required for normal morphology and precise division. This hypothesis suggests:

1. **Physical defaults + minimal molecular override** can support basic reproduction 2. **Additional molecular layers** are required for precision and morphological normality 3. The **threshold concept** is testable: systematic gene reduction should reveal where physical constraint dependence increases sharply

This balanced interpretation acknowledges that syn3.0 provides evidence for both physical constraint sufficiency (cells do divide) and molecular necessity (division is imprecise and abnormal).

Mycoplasma Diversity:

Wall-less mycoplasmas achieve cell cycles through diverse molecular adaptations:

• **FtsZ-dependent division**: Some mycoplasmas retain FtsZ and use conventional division machinery (Fisher et al., 2019; Sladek, 2019)

• **FtsZ-independent mechanisms**: Other mycoplasmas have evolved entirely different division strategies, including:

• **Septum formation with alternative cytoskeletal elements**

• **Budding-like division** in some species

• **Mechanical constriction** without FtsZ rings

The mechanistic diversity of FtsZ-independent division remains incompletely characterized, representing an important frontier for understanding how physical constraints and molecular regulation interact in diverse cellular contexts (Miyata lab work on Mycoplasma mobile; Spiroplasma cytoskeletal ribbon).

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6. Experimental Evidence and Stochasticity

6.1 In Vitro Reconstitution: Core Processes Self-Organize

Key Findings:

• Purified FtsZ forms dynamic patterns and Z-rings in liposomes (Osawa & Erickson, 2013; Ramirez-Diaz et al., 2021)

• Minimal systems can recapitulate aspects of division (García et al., 2021)

Limitations:

• Reconstituted systems lack the complexity of real cells

• Often require non-physiological conditions

• Self-organization doesn't equal physiological regulation

6.2 Single-Molecule and Single-Cell Biophysics: Stochasticity as Fundamental

Advanced Techniques:

• Optical tweezers, FRET, microfluidics, high-speed time-lapse microscopy (Joo et al., 2008; Wang et al., 2010; Norman et al., 2015; Neuman & Nagy, 2008)

• Magnetic tweezers for force measurement (Gosse & Croquette, 2002; Strick et al., 1996)

Key Findings:

• Mechanical forces affect molecular behaviors (Bisson-Filho et al., 2017)

• **Stochasticity has functional consequences**: Cell-to-cell variability in division timing and other processes is not simply meaningless noise but has functional significance for population survival and adaptation (Kiviet et al., 2014; Balaban et al., 2004; Levin et al., 2022; Synodinos et al., 2023)

• Individual cells show diverse behaviors within clonal populations

Cell-to-Cell Variability: An Integrated Perspective on Stochasticity

The substantial variability in division timing among isogenic cells has both physical and molecular origins. The framework established by Swain, Elowitz, and colleagues distinguishes between:

Intrinsic noise: Stochasticity arising from the random nature of biochemical reactions within individual cells, including:

• Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019)

• Brownian motion affecting molecular encounters (Parry et al., 2014)

• Stochastic gene expression and low-copy-number effects (Swain et al., 2002; Elowitz et al., 2002; Paulsson, 2004; Taniguchi et al., 2010)

• Random timing of key molecular events (Müller et al., 2019)

Extrinsic noise: Cell-to-cell variation arising from differences in the cellular environment or global state, including:

• Variations in growth rate, cell size, or cell cycle stage (Kiviet et al., 2014)

• Fluctuations in global parameters like ribosome concentration or metabolic state (Synodinos et al., 2023)

• Division errors and asymmetries that propagate to daughter cells (Balaban et al., 2004)

Physical origins of stochasticity:

• Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019)

• Brownian motion affecting molecular encounters (Parry et al., 2014)

Molecular origins of stochasticity:

• Stochastic gene expression (Raser & O'Shea, 2005; Kaern et al., 2005)

• Random timing of key molecular events (Müller et al., 2019)

• Low-copy-number effects for key regulators (Paulsson, 2004; Taniguchi et al., 2010)

Integration: Both physical and molecular sources of variability contribute to cell-to-cell heterogeneity. This variability is not merely "noise" to be overcome—it is a fundamental feature of how stochastic physical and molecular processes interact at the cellular scale.

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7. Synthesis: An Integrated Framework

7.1 The Core Thesis and Novelty Argument

What We Propose:

We propose a hierarchical framework for understanding bacterial cell cycle regulation:

Physical constraints create measurable behavioral tendencies within normal physiological ranges. Molecular systems have evolved to detect and override these tendencies when they would produce incorrect or suboptimal outcomes. This relationship is fundamentally asymmetric: molecular systems can read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Critical Distinction from Previous Work:

Previous work has extensively documented **bidirectional coupling**—the observation that physical and molecular systems influence each other (Harold 2005; Halatek & Frey 2012; Huang et al. 2013, 2019; Di Ventura & Sourjik 2022). This is well-established and uncontroversial.

Our claim is different: We propose that the **directionality of information flow** is asymmetric, even when the **causal influence** appears bidirectional. Specifically:

• **Physical → Molecular**: Physical states (cell size, geometry, DNA topology) are detected by molecular systems (e.g., Min system sensing geometry, topoisomerases sensing supercoiling)

• **Molecular → Physical**: Molecular systems actively intervene to alter physical states (e.g., FtsZ constricting the cell, topoisomerases altering DNA topology)

• **Physical → Molecular (reverse)**: Physical constraints do **not** detect or compensate for molecular states—they simply are what they are

What Makes This Distinction Operationalizable:

The key test is to examine what happens when one system is perturbed:

Perturbation Type	Bidirectional Coupling Prediction	Hierarchical Override Prediction
Physical perturbation (e.g., alter cell size using microfluidics)	Molecular systems compensate bidirectionally	Molecular systems compensate (they detect and override)
Molecular perturbation (e.g., deplete FtsZ)	Physical systems compensate bidirectionally	**No physical compensation occurs**—physical consequences follow passively

Critical Experimental Test:

Consider FtsZ depletion using a degron system:

• **Bidirectional coupling would predict**: As FtsZ levels drop, the cell would activate compensatory mechanisms that maintain division at the appropriate size, perhaps by altering turgor pressure or membrane mechanics to achieve the same physical outcome

• **Hierarchical override predicts**: As FtsZ levels drop, division simply fails to occur at the correct time. Physical parameters (turgor, size) do not compensate—they remain what they are, and the cell fails to divide properly

Evidence Supporting Hierarchical Organization:

1. **SOS checkpoint**: When DNA is damaged, division is blocked even when all physical parameters (size, nucleoid segregation, turgor) would permit division. Physical states do not compensate for this molecular decision.

2. **Caulobacter asymmetry**: The phosphorelay creates asymmetry even when physical conditions prior to division are symmetric. Physical states do not compensate to restore symmetry.

3. **SOS checkpoint**: When DNA damage occurs, SulA inhibits FtsZ polymerization, blocking division even when all physical parameters (size, nucleoid segregation, turgor) would permit division. Physical states do not compensate to maintain division.

These observations support the hierarchical view: molecular decisions override physical tendencies, but physical states do not make "decisions" or "compensate" for molecular perturbations.

Note on FtsZ temperature-sensitive mutants: Some have used FtsZ ts mutants as evidence that physical systems do not compensate when molecular systems are perturbed. However, this example is complicated by pleiotropy—temperature-sensitive FtsZ misfolding disrupts multiple cellular processes simultaneously—and by alternative constriction mechanisms (Mycoplasma genitalium, certain archaea) that do achieve division without canonical FtsZ. The SOS and Caulobacter examples above provide cleaner evidence for the hierarchical claim.

Our Contributions and Intellectual Context:

Contribution #1: Asymmetric Information Flow as a Distinct Claim

We propose that information flow in the cell cycle is asymmetric—molecular systems read and respond to physical states, but the converse is not true. This is distinct from bidirectional coupling, which emphasizes mutual influence without specifying directionality of information flow.

Contribution #2: Physical Tendencies as Measurable Quantities

Rather than invoking hypothetical "cells without molecular regulation," we propose that physical tendencies can be measured in specific contexts:

• **In minimal cells** (JCVI-syn3.0): Division shows high variability and strong correlation with physical parameters, suggesting physical tendencies dominate when molecular regulation is minimal

• **In specific mutants**: For example, cells lacking nucleoid occlusion show division through nucleoids at higher frequencies, revealing the underlying physical tendency of FtsZ to form rings independent of nucleoid position

• **Under stress conditions**: When molecular systems are compromised, physical tendencies become more apparent

Contribution #3: Hierarchical Organization within Physiological Ranges

We propose hierarchical organization where:

• **Foundation layer**: Physical constraints create behavioral tendencies

• **Override layers**: Molecular systems detect physical states and intervene when necessary

• **Key qualification**: This applies within physiological ranges. Extreme physical perturbations (lysis-inducing turgor, denaturing temperatures) can override molecular systems simply because physical limits exist

Connection to Prior Work:

Hierarchical biological organization has been discussed by Noble (2012) in the context of downward causation. Our contribution is to apply this specifically to the physical-molecular interface in bacterial cell cycle regulation, with explicit focus on asymmetric information flow.

Figure 1: Hierarchical Framework Schematic

``` ■ ■ ■ MOLECULAR OVERRIDE LAYERS ■ ■ ■ ■ ■ ■ ■ ■ Checkpoint ■ ■ Developmental ■ ■ Precision ■ ■ ■ ■ Systems ■ ■ Switches ■ ■ Timing ■ ■ ■ ■ (SOS, etc.) ■ ■ (Caulobacter) ■ ■ Systems ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ▼■ ■ ■ ■ ■ MOLECULAR ■ ■ ■ ■ ■ SENSING & ■ ■ ■ ■ ■ INTERVENTION ■ ■ ■ ■ ■ LAYER ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ▼ ■ ■ ■ ■ PHYSICAL FOUNDATION LAYER ■ ■ ■ ■ ■ ■ ■ ■ Turgor ■ ■ Nucleoid ■ ■ DNA ■ ■ ■ ■ Pressure ■ ■ Geometry ■ ■ Topology ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ Physical constraints create measurable tendencies ■ ■ (size-dependent division probability, entropic ■ ■ chromosome segregation, crowding-dependent rates) ■ ■ ■

INFORMATION FLOW

Physical → Molecular: Molecular systems detect physical states Molecular → Physical: Molecular systems intervene to override Physical → Molecular (reverse): Does NOT occur - no physical "detection" ```

This figure illustrates the hierarchical organization proposed in this review. Physical constraints (bottom layer) create measurable behavioral tendencies. Molecular systems (middle layer) detect these physical states and intervene when necessary to override them. Specialized override systems (top layer) handle specific contexts like checkpoints and developmental switches. The arrows show asymmetric information flow: molecular systems read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

**Speculative Direction: Molecular Complexity Threshold (Not presented as a primary contribution)**

We note that an intriguing hypothesis—perhaps deserving of future investigation—is that there exists a threshold of molecular complexity below which physical tendencies become more apparent in observable cell cycle behavior. JCVI-syn3.0 (473 genes) divides but with abnormal morphology and high variability. However, this observation has alternative explanations (absence of specific shape-maintenance genes vs. exposure of physical defaults), and without a quantitative metric for "molecular complexity," the threshold concept remains verbal rather than quantitative. We
```

present this as a direction for future research rather than a current contribution.

Distinguishing Our Framework from Previous Work:

Aspect	Previous Framing	Our Framing
Information flow	Bidirectional (mutual influence)	**Asymmetric** (molecular reads physical, not vice versa)
Physical role	Constraints on molecular systems	**Foundation** creating measurable tendencies
Molecular role	Control systems in coupled network	**Override systems** detecting and intervening
Key test	Both directions show compensation	**Only molecular → physical shows compensation**

Why This Distinction Matters:

1. **Predictive power**: The asymmetric flow prediction can be tested experimentally by perturbing molecular systems and asking whether physical parameters compensate (they shouldn't)

2. **Interpretive framework**: Minimal cell data make sense as systems operating near physical default regimes

3. **Evolutionary implications**: Suggests molecular regulation evolved as layers added to physical foundations, not as parallel systems from the beginning

2. **Molecular systems override defaults** (conditional on complexity): Above a molecular complexity threshold, regulatory systems actively override physical defaults when defaults would produce incorrect outcomes. The **"dominance" of molecular regulation is context-dependent**:

• **Strong molecular dominance**: Checkpoint responses (SOS), developmental switches (Caulobacter), precision timing (multifork replication)

• **Weak molecular dominance**: Basic division under normal growth conditions (some molecular input but physical defaults contribute)

• **Physical dominance**: Minimal cells, stress conditions, molecular perturbations

3. **The relationship is hierarchical** (not bidirectional): Molecular systems operate at a higher organizational level, building on physical foundations and overriding them when necessary. Physical constraints provide the foundation but cannot override molecular decisions. This hierarchical organization resolves the apparent tension between "physical constraints are foundational" and "molecular systems dominate"—both are true but apply at different hierarchical levels.

4. **Evolution adds hierarchical layers** (not symmetric coupling): Evolution proceeds by adding molecular override layers on top of physical foundations, not by creating balanced bidirectional coupling.

5. **Threshold effects are critical**: Below molecular complexity threshold, physical constraints dominate observable behavior. Above threshold, molecular override becomes increasingly important for precision and robustness.

6. **Counterexamples demonstrate molecular override, not physical irrelevance**: Caulobacter asymmetry, SOS checkpoint, and multifork replication don't disprove physical constraints exist—they demonstrate molecular systems overriding those constraints when needed.

7. **Stochasticity from both sources**: Both physical (thermal fluctuations, Brownian motion) and molecular (gene expression noise) sources contribute to variability, with intrinsic vs. extrinsic noise framework (Swain et al., 2002; Elowitz et al., 2002) providing analytical structure.

7.2 The Min System: A Case Study in Hierarchical Integration

The Min oscillation system provides an illuminating example of how molecular systems interface with physical constraints:

Physical constraint: Cell geometry (length, curvature, pole position)

Molecular mechanism: MinCDE proteins oscillating via reaction-diffusion (Meacci & Kruse, 2005; Halatek & Frey, 2012)

Coupling mechanism (with active debate):

• Min oscillation period correlates with cell length (Di Ventura & Sourjik, 2022; Lutz et al., 2023)

• **Active geometric sensing hypothesis**: Min system actively "reads" cell geometry and adjusts behavior accordingly

• **Passive consequence hypothesis**: Oscillation period adaptation emerges from reaction-diffusion dynamics operating in larger volumes without explicit sensing

• **Current status**: The field actively debates these alternatives (Ramm et al., 2019; Di Ventura & Sourjik, 2022; Lutz et al., 2023)

Regardless of mechanism, the Min system demonstrates molecular-level encoding of geometric information, providing precise division site positioning that goes beyond what physical geometry alone would achieve.

Why this matters for our framework:

• Demonstrates molecular systems achieving spatial precision beyond physical defaults

• Shows how molecular regulation interfaces with physical constraints (whether through active sensing or passive coupling)

• Provides a concrete example of hierarchical organization: physics creates geometry-dependent possibilities, molecular system selects optimal division site

Figure 2: Min System Hierarchical Integration

``` PHYSICAL CONSTRAINT Cell Geometry (Length) ■ ▼ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ MIN OSCILLATION SYSTEM (Molecular) ■ ■ ■ ■ Pole A Pole B ■ ■ ●■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■● ■ ■ ■ ■ ■ ■ MinE■←■■■■■■ MinD ■■■■■→■MinE ■ ■ ■ (ATP-dependent) ■ ■ ■ ▼ ▼ ■ ■ MinC (inhibitor) ←■■■■■■■■■■ ■ ■ ■ ■ OSCILLATION DYNAMICS: ■ ■ MinD-ATP binds membrane → recruits MinC → inhibits Z-ring ■ ■ MinE stimulates MinD-ATP hydrolysis → releases MinD ■ ■ Result: Pole-to-pole oscillation of MinC inhibition zone ■ ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■ ■ ▼ MOLECULAR OVERRIDE Z-ring forms at midcell (low MinC) ■ ▼ FUNCTIONAL OUTCOME Precise division site positioning

INFORMATION FLOW IN MIN SYSTEM:

Physical → Molecular: Cell geometry (length) influences Min oscillation period → longer cells have longer oscillation periods

Molecular → Physical: MinC inhibition pattern determines where Z-ring forms → overrides uniform Z-ring formation that physical geometry alone would produce

Key Question: Does Min "actively sense" geometry or "passively respond" to changing volume? Current status: DEBATED ```

This figure illustrates how the Min system interfaces with physical constraints. The physical constraint (cell geometry/length) influences the molecular oscillation system (MinCDE). The molecular system then overrides physical defaults by creating a precise division site at midcell that would not occur based on physical geometry alone. The active vs. passive geometric sensing question remains experimentally unresolved.

**Open questions**: Distinguishing active sensing from passive consequence remains experimentally challenging. Future work combining microfluidic geometry manipulation with single-molecule imaging may resolve this debate.
```

7.3 Testable Predictions

Our hierarchical framework makes several testable predictions:

1. **Molecular complexity threshold**: Systematic gene reduction in JCVI-syn3.0 or other minimal cells should reveal a threshold below which division becomes increasingly imprecise and variable, reflecting dominance of physical defaults above the threshold and increasing reliance on molecular override below it.

2. **Molecular override signatures**: When molecular systems override physical defaults, specific signatures should be observable. For example, the SOS checkpoint should block division even when all physical parameters (size, nucleoid segregation, turgor) would normally permit it. Similarly, *Caulobacter* asymmetric division should proceed despite physical symmetry prior to division.

3. **Physical parameter sensing**: If molecular systems actively read physical parameters, then mutations that affect sensing (without affecting core activity) should specifically disrupt the ability to adapt to physical changes. Example: Min system mutants that cannot properly read cell length should fail to adapt oscillation period to cell size, without affecting oscillation dynamics per se.

4. **Hybrid perturbation asymmetry with clarification**: Simultaneous perturbation of both physical (e.g., osmotic conditions) and molecular (e.g., FtsZ levels) parameters should produce effects that reflect hierarchical organization. Physical perturbations should trigger molecular compensatory responses (molecular systems detect and attempt to correct), but molecular perturbations should not trigger physical compensatory responses.

Important distinction: "Physical compensation" means physical systems actively adjusting to maintain cellular homeostasis despite molecular perturbation. "Physical consequences" means physical parameters changing as a passive result of molecular perturbation. The hierarchical framework predicts physical consequences (e.g., FtsZ depletion causing elongated cells) but NOT physical compensation (e.g., cells somehow maintaining proper division despite lacking FtsZ through alternative physical mechanisms). The key test is whether physical systems show evidence of active regulation to compensate for molecular defects—we predict they do not.

5. **Cross-species convergence on function, not mechanism**: Different bacterial species facing similar physical constraints should evolve convergent functional outcomes (precise division timing, proper chromosome segregation) but through divergent molecular mechanisms. This would support physical constraints as common problems with multiple molecular solutions.

6. **Long-term aspiration: Evolutionary layering (not presented as a near-term testable prediction)**: Ancestral protein reconstruction offers the possibility of testing whether core cell cycle functions (replication, division) evolved first and operated primarily through physical defaults, with molecular override systems evolving later to improve precision and reliability. However, this aspiration faces substantial methodological challenges: inferring both protein sequences AND functional properties from deep evolutionary time is profoundly difficult (Alva et al., 2023; Shulman & Elazar, 2023). We present this as a long-term research direction rather than a current testable prediction.

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8. Implications and Future Directions

8.1 For Origins of Life Research

Implications:

• Physical constraints likely provided some organizational logic in early protocells

• **However**: Early protocells likely had **imprecise, inefficient** cell cycles with substantial variability

• Evolution of molecular regulation was essential for achieving the precision observed in modern bacteria

8.2 For Synthetic Biology

Design Principles:

• Understand physical constraints before designing molecular systems

• **But**: Recognize that molecular sophistication is often required for precision

• Match regulatory complexity to functional requirements

• Consider hybrid approaches that exploit both physical defaults and molecular control

8.3 For Fundamental Cell Biology

Key Questions:

• Which aspects of cell cycle regulation are **physically constrained** vs. **contingent molecular solutions**?

• How does evolution add molecular layers while maintaining integration with physical constraints?

• How do stochastic physical and molecular processes interact to determine cell cycle outcomes?

8.4 For Broader Biological Theory

Our hierarchical framework connects to broader questions in systems biology and philosophy of science. Cell cycle regulation cannot be reduced to either molecular or physical explanations alone (Noble, 2012; Bizzarri et al., 2013). Our framework's emphasis on asymmetric causal relationships—molecular systems can override physical defaults but not vice versa—challenges simple hierarchical causal models (Pearl, 2009; Woodward, 2003). Rather than claiming that physical constraints are "sufficient" or "insufficient," we specify when each dominates based on molecular complexity threshold and functional context. This moves beyond the question of "what counts as explanation in biology" (Weisberg, 2007) to specifying conditions under which different types of explanations apply.

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9. Future Experimental Directions

9.1 Critical Experimental Tests

1. Simultaneous Physical-Molecular Perturbation Systematically vary both physical parameters (crowding agents, confinement, membrane tension) and molecular components (gene knockouts, protein overexpression) while measuring effects on cell cycle precision, robustness, and stochasticity.

2. Evolutionary Reconstruction and Functional Testing Use ancestral sequence reconstruction to infer sequences of key cell cycle regulators, then express ancestral proteins in modern cells to assess functionality (Alva et al., 2023; Shulman & Elazar, 2023).

3. Stochastic Signature Analysis Single-cell tracking to distinguish molecular vs. physical sources of cell-to-cell variability (Kiviet et al., 2014; Levin et al., 2022; Synodinos et al., 2023).

4. Hybrid In Vitro Reconstitution Develop in vitro systems combining physical constraints (crowding agents, confinement, membrane curvature) with molecular components (García et al., 2021).

5. Comparative Analysis Across Bacteria Systematically compare cell cycle regulation across diverse bacterial species, including:

• Gram-positive vs. Gram-negative (different turgor pressures)

• Species with and without cell walls (mycoplasmas)

• Halophiles and other extremophiles

• Different division mechanisms (FtsZ-based vs. alternatives)

This comparative approach would test whether the integrated framework generalizes across diverse physical contexts and whether different molecular solutions have evolved to manage different physical constraints.

9.2 Computational and Theoretical Approaches

Multi-scale Modeling: Develop models that explicitly integrate physical constraints and molecular regulation (Deng et al., 2020; Huang et al., 2019; Shi et al., 2018). Current models typically focus on either physical or molecular aspects; integrated models could reveal emergent properties.

Causal Inference: Apply rigorous causal discovery algorithms to experimental data to distinguish physical from molecular causation:

• PC algorithm and extensions for causal structure learning (Spirtes et al., 2000)

• Intervention analysis using do-calculus (Pearl, 2009)

• Counterfactual reasoning for evaluating what would happen under different conditions

Comparative Genomics: Systematic comparison of cell cycle regulatory architectures across bacterial species could reveal:

• Conservation of physical constraint management strategies

• Convergent evolution of molecular solutions to similar physical constraints

• Correlation between environmental physical constraints and regulatory complexity

Single-Cell Data Analysis: Leverage advances in single-cell technologies:

• Microfluidics for long-term single-cell tracking (Wang et al., 2010)

• Time-lapse microscopy for measuring division timing variability

• Integration with causal inference to identify regulatory relationships

Evolutionary Simulations: Simulate evolution of cell cycle regulation under different physical constraints to understand when different levels of molecular sophistication evolve (using agent-based models or population genetics simulations).

9.3 Open Questions

Fundamental Questions:

• What determines when molecular systems override vs. work within physical constraints?

• How did integrated physical-molecular regulation evolve?

• What is the minimal level of integration required for functional cell cycles?

• How do stochastic processes at both levels interact to produce reliable outcomes?

Technical Challenges:

• Measuring physical parameters in vivo with sufficient precision

• Manipulating physical parameters independently of molecular components

• Distinguishing molecular override of physical defaults from parallel coupling mechanisms

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10. Conclusion

The bacterial cell cycle operates at the interface of physics and biology. Physical constraints—turgor pressure, nucleoid geometry, DNA topology, macromolecular crowding, entropic forces, and metabolic state—create measurable behavioral tendencies that become particularly apparent when molecular regulation is minimal, compromised, or absent in specific regulatory contexts. Molecular regulation has evolved to detect and override these physical tendencies when they would produce incorrect or suboptimal outcomes.

The evidence supports a **hierarchical framework** where: 1. Physical constraints create measurable behavioral tendencies observable in specific contexts 2. Molecular systems have evolved to detect and override these tendencies when necessary 3. The relationship is fundamentally asymmetric: molecular systems can read and respond to physical states, but physical constraints do not "read and respond" to molecular states 4. Evolution has layered molecular sophistication on top of physical foundations 5. Key examples (*Caulobacter*, SOS checkpoint, multifork replication) demonstrate molecular override of physical tendencies 6. Stochasticity from both physical and molecular sources contributes to variability

Key Contributions of This Review:

1. **Hierarchical framework with asymmetric information flow**: We propose that molecular systems read and respond to physical states, but physical constraints do not read and respond to molecular states—distinct from bidirectional coupling views which emphasize mutual influence without specifying directional information flow

2. **Measurable physical tendencies**: Rather than hypothetical "cells without regulation," we propose that physical tendencies can be measured in minimal cells (JCVI-syn3.0), specific mutants, and stress conditions where molecular regulation is compromised

3. **Testable distinction**: The hierarchical framework makes testable predictions—for example, molecular perturbations should not trigger physical compensation, whereas physical perturbations should trigger molecular compensatory responses

4. **Honest assessment of counterexamples**: We explicitly address cases where molecular regulation clearly dominates and explain how these fit within the hierarchical framework

5. **Critical evaluation of evidence**: We distinguish between well-established mechanisms and speculative areas, particularly regarding geometric sensing

6. **Min system as case study**: We highlight the Min oscillation system as a paradigmatic example of molecular systems interfacing with physical constraints, with honest acknowledgment of ongoing debates

7. **Bacterial diversity**: We acknowledge that our hierarchical framework must account for diverse bacterial species with vastly different physical constraints. Gram-positive bacteria such as *B. subtilis* experience higher turgor pressures than Gram-negative bacteria such as *E. coli*, though estimates vary widely (Whatmore & Reed, 1990; Koch, 1983). All achieve functional cell cycles through adapted molecular regulation—supporting the view that diverse molecular solutions evolve to manage different physical contexts.

Final Synthesis:

The most exciting future work will elucidate how molecular systems have evolved to detect and override physical tendencies. The Min system exemplifies a general design principle: successful biological systems achieve their remarkable capabilities by evolving molecular interfaces that can detect physical states and intervene when necessary. This principle likely extends throughout all living systems.

The hierarchical framework we propose connects to broader questions in systems biology, origins of life research, and the philosophy of biology. It suggests that early protocells likely exhibited physical tendency-dominated behavior (imprecise but functional), with evolution adding successive layers of molecular override to achieve the precision and reliability observed in modern bacteria. A complete theory of cellular organization must recognize this hierarchical layering, from physical foundations through molecular detection and override systems to cellular outcomes.

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[DUPLICATE ENTRY REMOVED - This paper is correctly cited as Le Gall et al. (2022)]

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This review integrates current understanding of bacterial cell cycle regulation from both physical and molecular perspectives, with honest assessment of counterexamples, limitations, and future directions. References have been manually verified against primary sources; any remaining errors are the responsibility of the authors.

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